POWER PLANT ENGINEERING

INTRODUCTION

POWER

- Definition of Power

Power is the time rate at which work is done or energy is transferred. In calculus terms, power is the derivative of work with respect to time.

Work= Force X Distance

= FXd

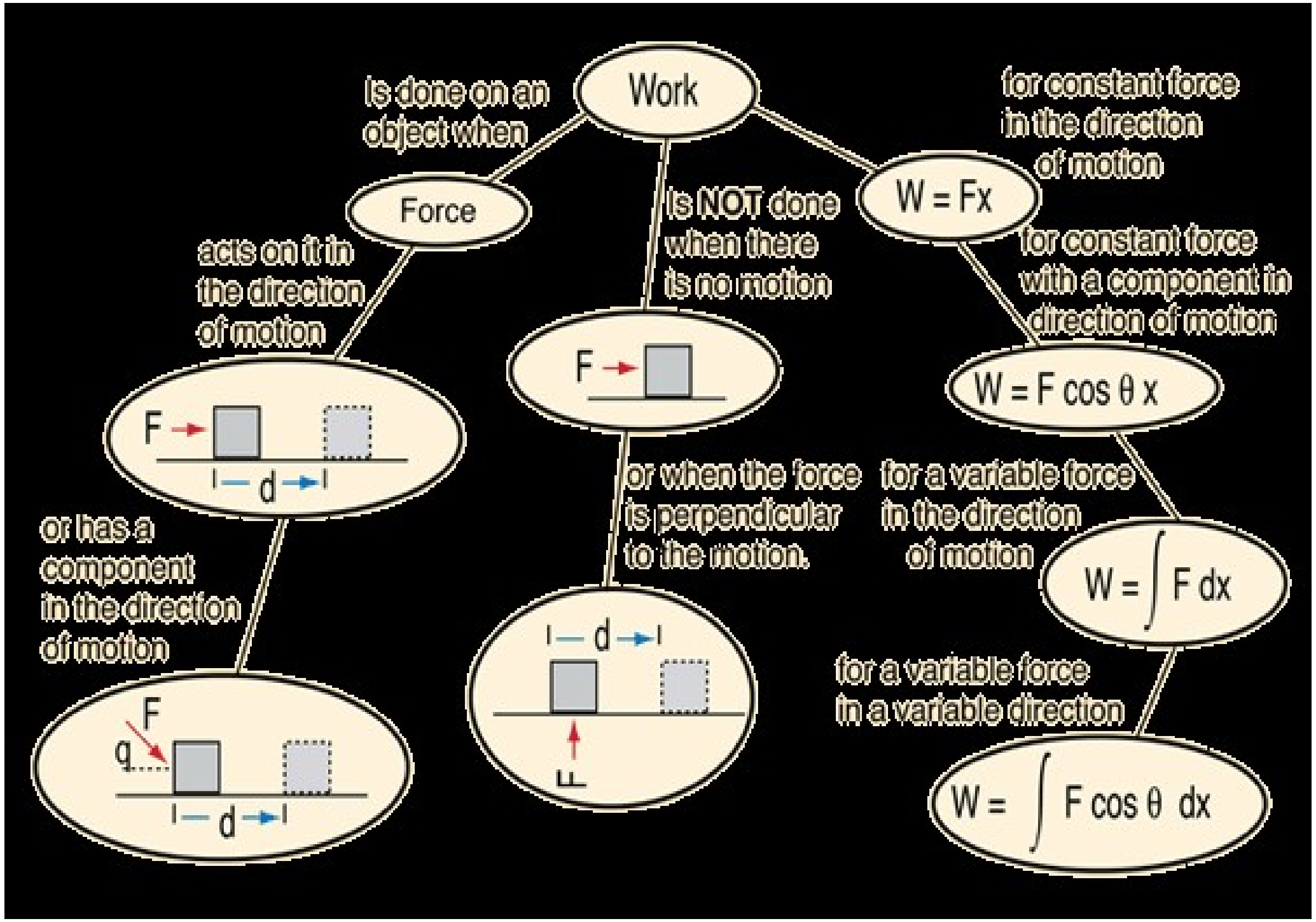
Unit of work in SI System is Joule (N-m)

Power=work/time=N-m/s=watt

ENERGY

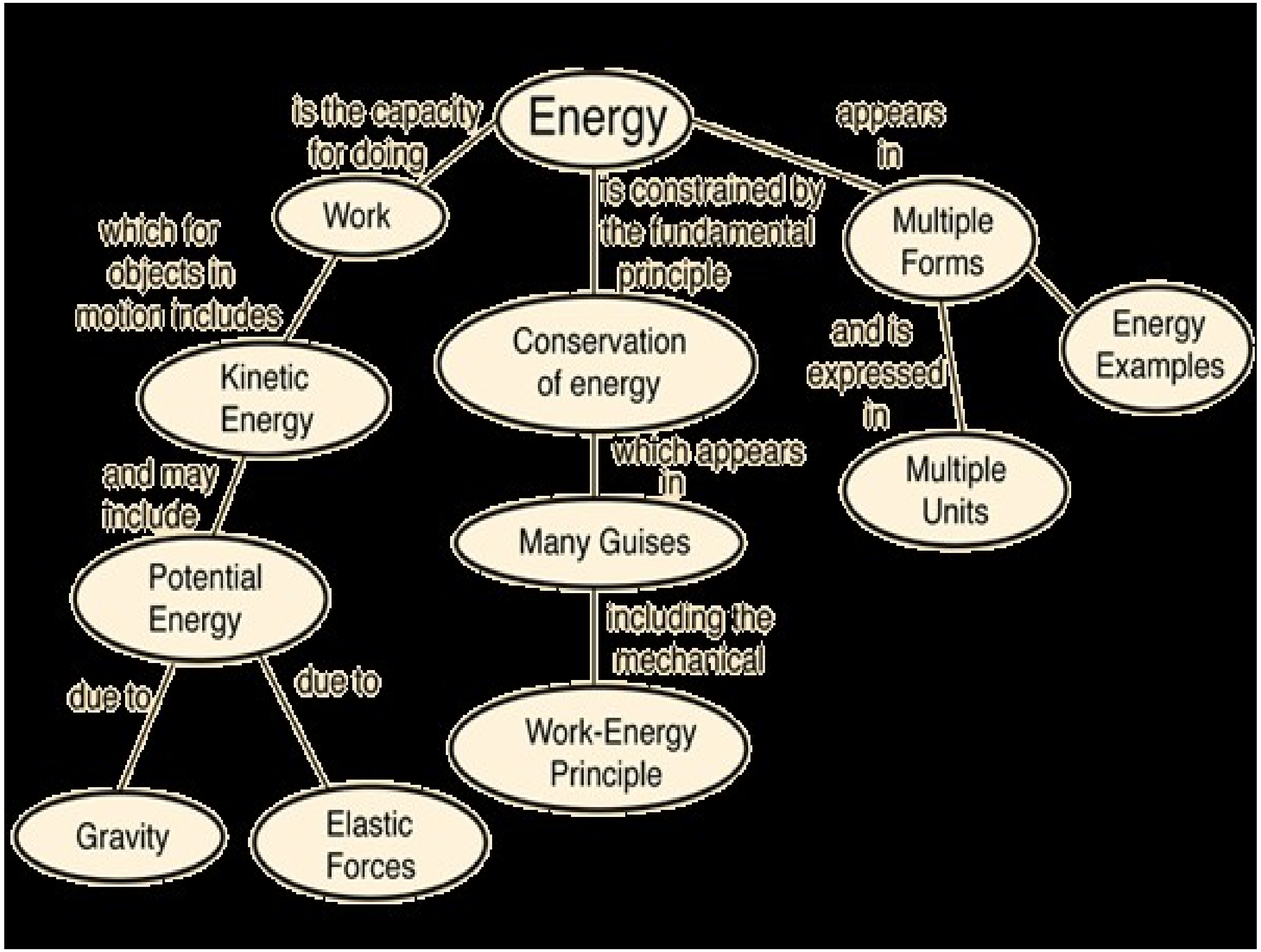
- Capacity to do work
- Unit of Energy is the same as unit of work
- In SI system-Joule

WORK



Work
is done on an object when
Force
acts on it in the direction of motion
F
d
or has a component in the direction of motion
F
q
d
is NOT done when there is no motion
F
or when the force is perpendicular to the motion.
d
F
W = Fx
for constant force in the direction of motion
for constant force with a component in direction of motion
W = F cos θ x
for a variable force in the direction of motion
W = ∫ F dx
for a variable force in a variable direction
W = ∫ F cos θ dx

ENERGY



Energy
is the capacity for doing
Work
which for objects in motion includes
Kinetic Energy
and may include
Potential Energy
due to
Gravity
due to
Elastic Forces
is constrained by the fundamental principle
Conservation of energy
which appears in
Many Guises
including the mechanical
Work-Energy Principle
appears in
Multiple Forms
Energy Examples
and is expressed in
Multiple Units

WORK, ENERGY AND POWER

- Work-refers to an activity involving a force and movement in the direction of the force. A force of 20 N pushing an object 5 meters in the direction of the force does 100 joules of work.

- Energy-is the capacity for doing work. You must have energy to accomplish work - it is like the "currency" for performing work. To do 100 joules of work, you must expend 100

Power-2

- Units of Power
- Power=Work Done/time
- W=F X Distance/time
- W=N-M/s=J/s=Watt
- One KW=1000W
- One MW=1000KW
- Horsepower is a unit of power in the British system of measurement.
- One HP=746(745.699872 W) watts, assuming that a horse can move a mass of 746 KG to one meter in one second or 1492 KG to one meter in 2 secs. Usain bolt can run 100 min 9.58 secs. It is calculated that he expends 81.58 KJ of Energy during the run.
- Trained athletes can develop upto 2.5 HP.

Power-3 History of the Unit

- Development of steam engine(1770) necessitated the comparison with power generated by horses

33000 lbs-ft/min=one hp(optimistic)(A marketing tool)=550 ft·lbf/s

Or given that 1 hp = 550 ft·lbf/s, 1 ft = 0.3048 m, 1 lbf ≈ 4.448 N, 1 J = 1 N·m, 1 W = 1 J/s: 1 hp = 746 W

$$\begin{aligned}
1\,\mathrm{lbf} &= 1\,\mathrm{lb} \times g_\mathrm{n} \\
&= 1\,\mathrm{lb} \times 9.80665\,\tfrac{\mathrm{m}}{\mathrm{s}^2} / 0.3048\,\tfrac{\mathrm{m}}{\mathrm{ft}} \\
&\approx 1\,\mathrm{lb} \times 32.174049\,\tfrac{\mathrm{ft}}{\mathrm{s}^2} \\
&\approx 32.174049\,\tfrac{\mathrm{ft{\cdot}lb}}{\mathrm{s}^2} \\
1\,\mathrm{lbf} &= 1\,\mathrm{lb} \times 0.45359237\,\tfrac{\mathrm{kg}}{\mathrm{lb}} \times g_\mathrm{n} \\
&= 0.45359237\,\mathrm{kg} \times 9.80665\,\tfrac{\mathrm{m}}{\mathrm{s}^2} \\
&= 4.4482216152605\,\mathrm{N}
\end{aligned}$$

Units of power

- The following definitions have been widely used:
- Mechanical horsepower
- hp(I) ≡ 33,000 ft-lbf/min

- = 550 ft·lbf/s
- = 745.699872 W
- Metric horsepower
- hp(M) ≡ 75 kgf·m/s

- ≡ 735.49875 W
- Electrical horsepower
- hp(E) ≡ 746 W
- Boiler horsepower
- hp(S) ≡ 33,475 BTU/h

- = 9,809.5 W
- Hydraulic horsepower = flow rate (US gal/min) × pressure (psi) × 7/12,000

- or
- = flow rate (US gal/min) × pressure (psi) / 1714
- = 550 ft·lbf/s
- = 745.699872 W

Power & Energy-1

- Calorie-the amount of energy required to warm one gram water from by one $^{\circ}$C at standard atmospheric pressure.(One Kcal=One Kg by One deg)
- One Kcal=4.2 KJ
- Calorific value of diesel =10800cal/gm=45 KJ/gm
- To move a bus(16 tone) by one meter considering efficiency 30-35 % -one gm of diesel is required

Electrical Energy-1

- Difference between KW,KWh & KVA
- KW is unit of power
- KWh is unit of Energy(Calorie,Joule ,wh etc are units of energy)
- Energy=Power X Time
- Unit of Cost Electrical energy sold is Rs/KWHr or simply Rs/unit-which is as low as Rs4 /unit in Kerala
- To pump water using a 1 HP motor to fill a 1000 ltr capacity tank from a depth of 10 meters ,what is the cost & time?

Electrical Energy-2

- Total work done= 1000X10X9.8 N-M=98000 N-M
- Time taken=98000/746=131 secs
- Considering efficiency of pump at 75% and power factor of .65 for the pump,time to be taken=268 secs=4.47 approx 5mins=.083 hr
- Cost= .746X.083=0.062 Kwhr
- Cost of 1 unit=Rs5=500 paise
- Cost of .062 units=500X.062=31 paise

KW & KVA

- The capacity of generators are indicated as KVA
- Power factor
- Demand
- or example, to get 1 kW of real power, if the power factor is unity, 1 kVA of apparent power needs to be transferred (1 kW ÷ 1 = 1 kVA). At low values of power factor, more apparent power needs to be transferred to get the same real power. To get 1 kW of real power at 0.2 power factor, 5 kVA of apparent power needs to be transferred (1 kW ÷ 0.2 = 5 kVA). This apparent power must be produced and transmitted to the load in the conventional fashion, and is subject to the usual distributed losses in production.

Power plants

- Eg- 100 cc bike
- How its capacity usual specified
- Bajaj Discover DTS-Si 100 specifications are
- 94.38cc engine, 7.5 BHP of power @ 7500 rpm and 7.85 Nm @ 5000 rpm
- $BHP=2\prod NT/60$
- Lower gears more torque than power(A 2:1 gearbox halves BHP and doubles Torque, a 3:1 thirds and trebles, a 1:2 doubles and halves)

Torque

- Torque, moment or moment of force (see the terminology below), is the tendency of a force to rotate an object about an axis, fulcrum, or pivot. Just as a force is a push or a pull, a torque can be thought of as a twist to an object. Mathematically, torque is defined as the product of force and the lever-arm distance, which tends to produce rotation
- HP=TorqueXRPM/5252

Power Plant Capacities

- Common I C engine generators
- 5KVA to 2.8 KVA Honda Genset
- Petrol/Kerosene/ LPG
- 7 KW to 16000KW caterpillar Diesel Genset

Power Capacities of Devices-1

- 60W light bulb - 60W
- 100W light bulb - 100W
- Microwave : 600-1500W
- Washing Machine : 500W
- Vacuum Cleaner : 200-700W
- Iron : 1000W
- Ceiling Fan : 50-60W
- Laptop Computer : 20-50W
- Desktop Computer : 80-150W
- TV (19" colour) : 70W
- Clock radio : 1W
- Fridge : 500W
- 25" colour TV : 150W
- Over head water pump-375 to 750 W
- Thermal radiation emitted by a human body- 100W

Power Capacities of Devices-2

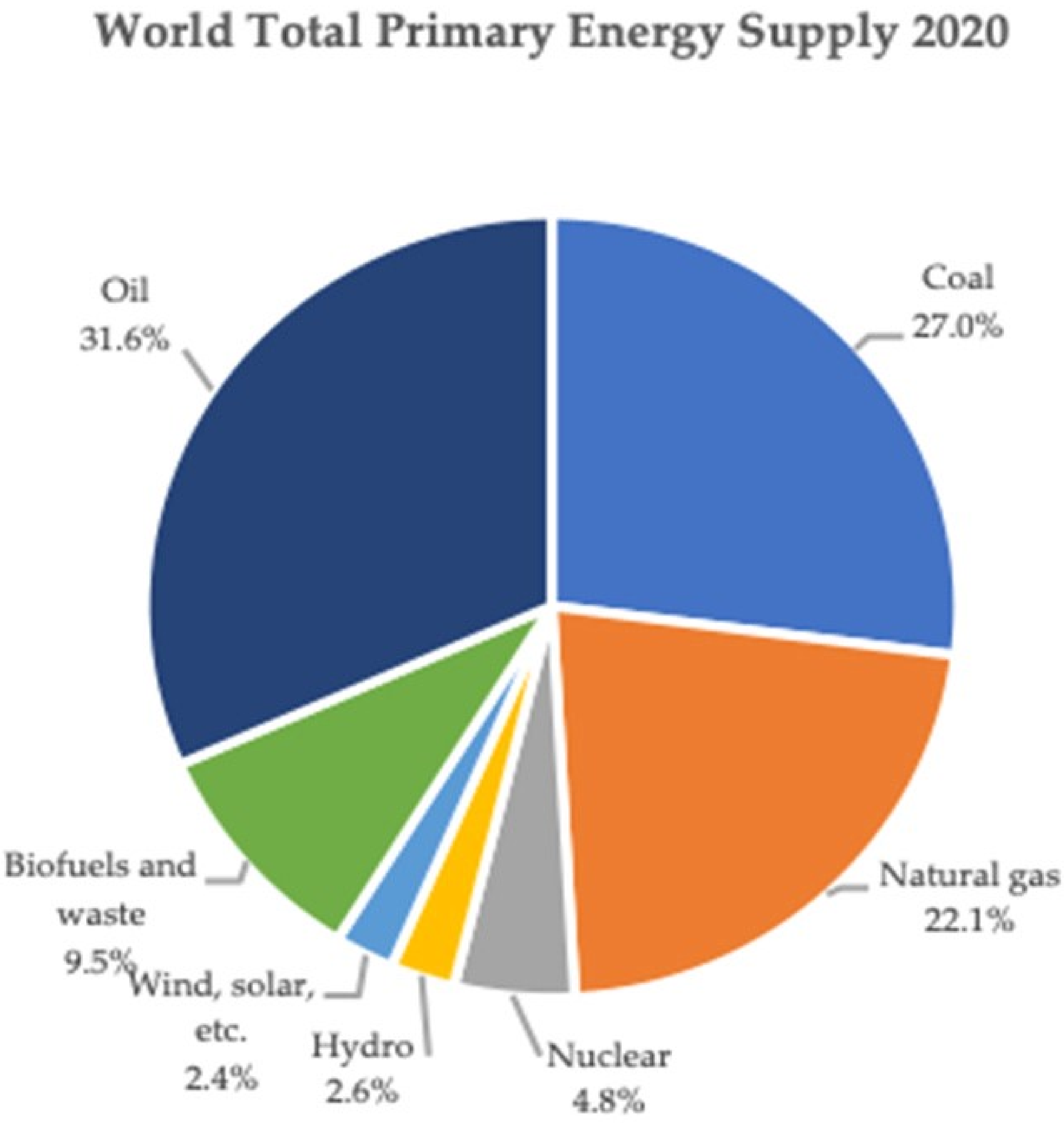
- Mobile Phone=1-3 W
- Usb drive .5-2 W
- Maximum output from a UMTS/3G mobile - 125 mW
- Bluetooth Class 2 radio, 10 m range-2.5mW
- Typical Wireless LAN transmission power in laptops 32 mW

Power Capacities of Devices-1

- Bike-7-10 BHP
- Car50-120 BHP
- Bus-160-200 BHP
- Train Engine-3000-7000 HP
- Cruise Ship-50000-125000 BHP
- Nuclear Submarine-50000-80000 HP

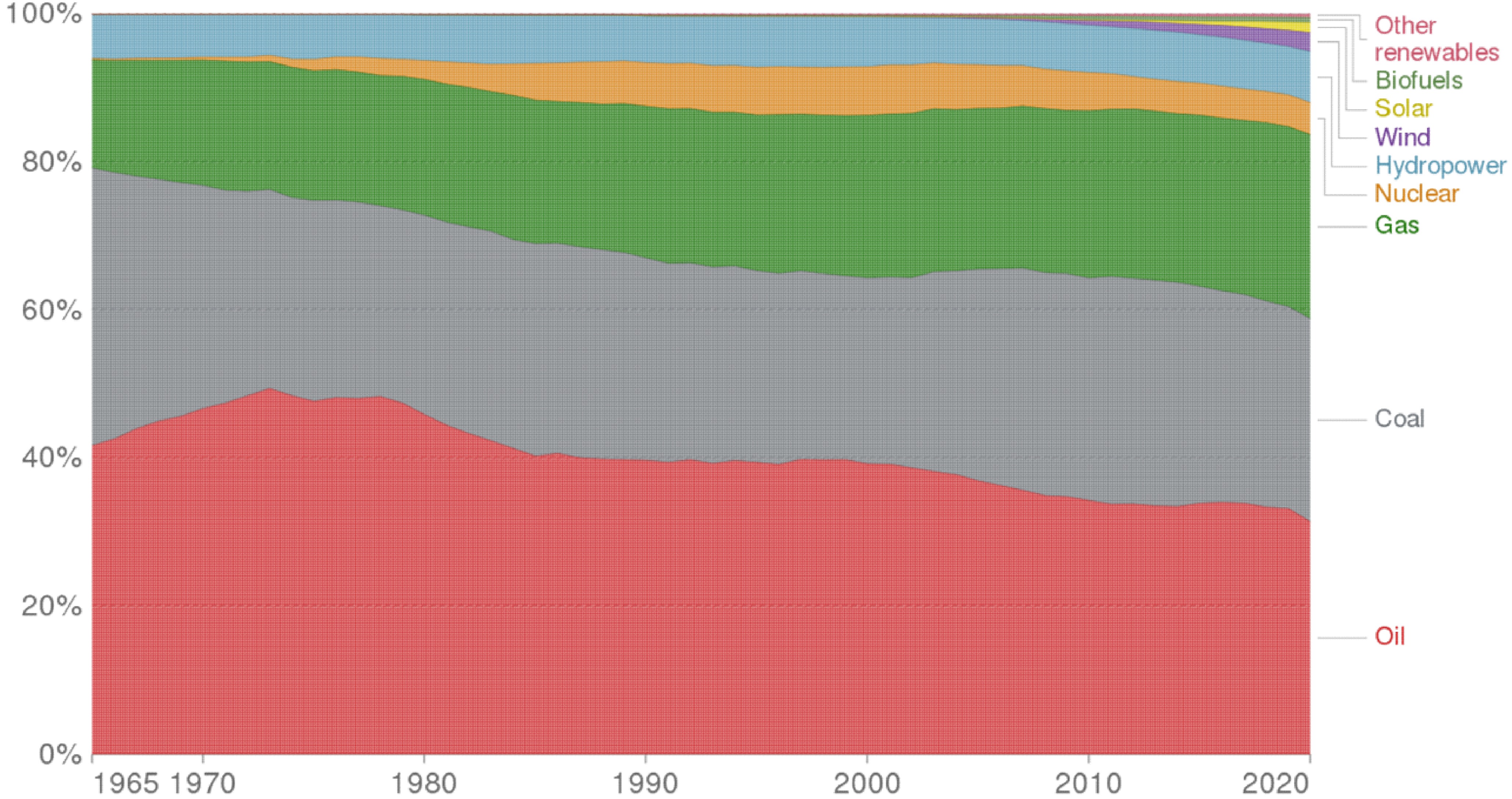
Energy Requirement World

- According to calculations by the World Energy Council (WEC), the world-wide energy consumption currently is about 14 billion oil equivalent per year .
- (One Litre of Oil=4.2 MJ)
- One Ton oil=1 GJ(Approx)


World Total Primary Energy Supply 2020
Oil
31.6%
Coal
27.0%
Natural gas
22.1%
Biofuels and
waste
9.5%
Wind, solar,
etc.
2.4%
Hydro
2.6%
Nuclear
4.8%

Energy consumption by source, World

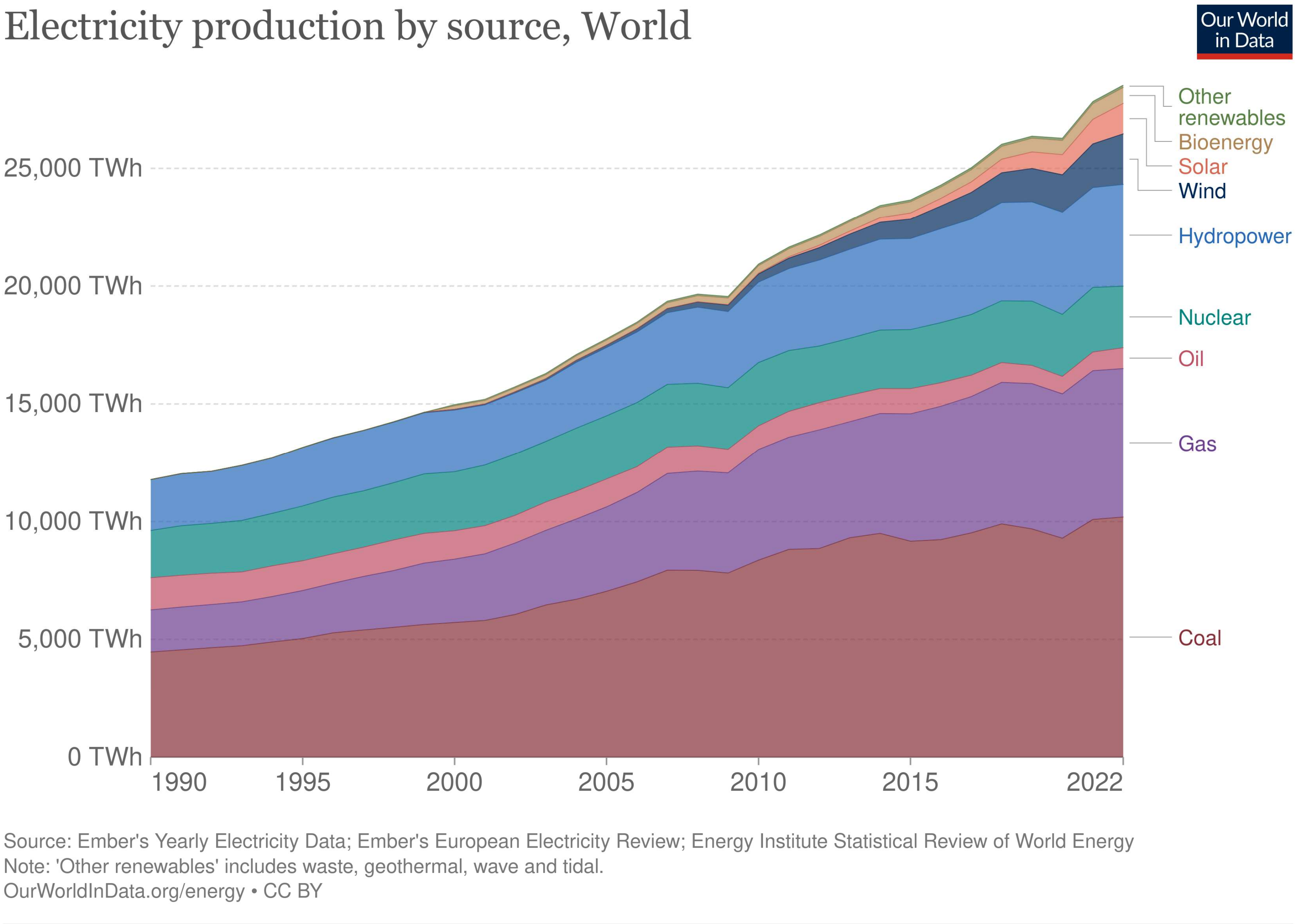
Primary energy consumption is measured in terawatt-hours (TWh). Here an inefficiency factor (the 'substitution' method) has been applied for fossil fuels, meaning the shares by each energy source give a better approximation of final energy consumption.



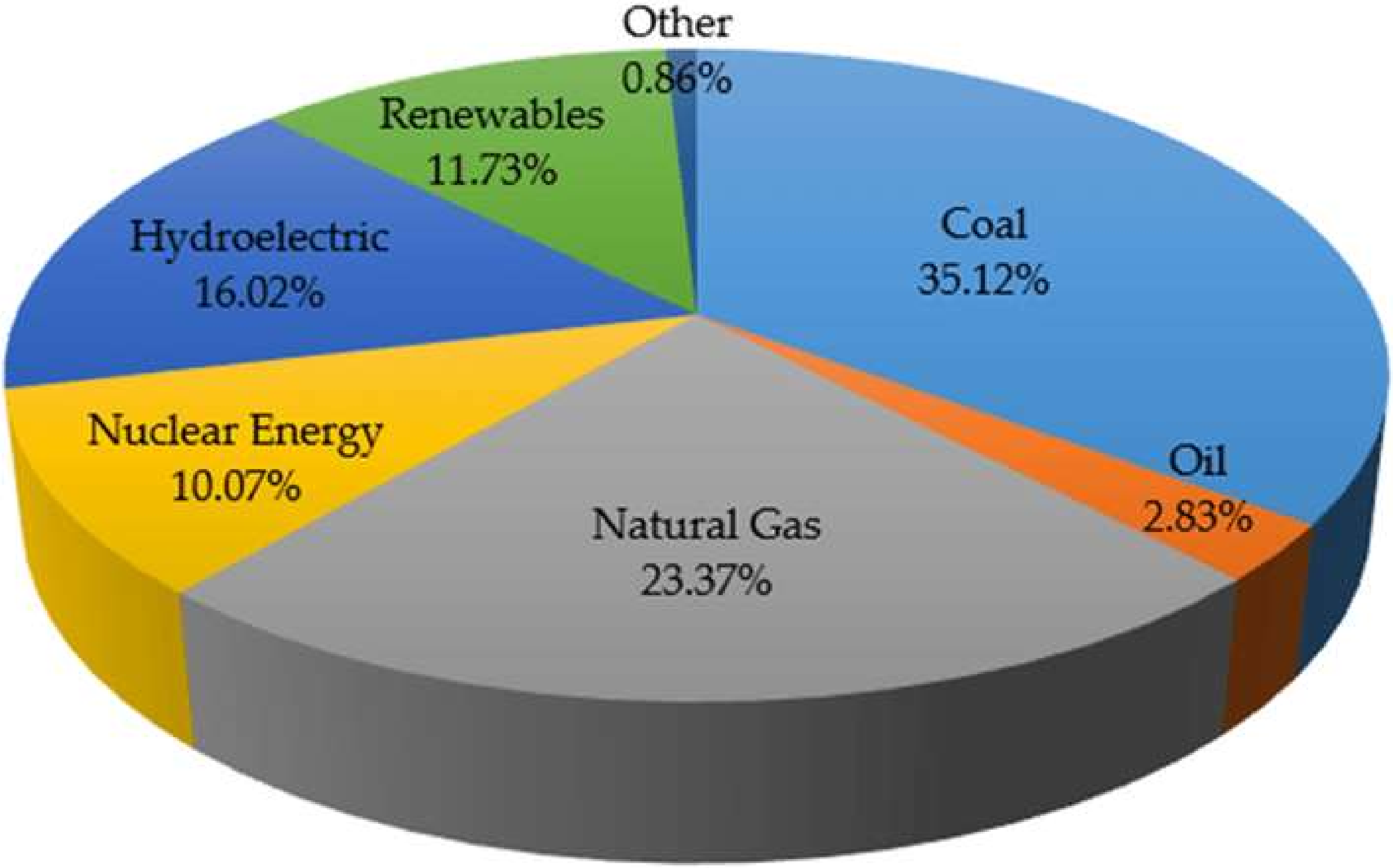
Source: BP Statistical Review of World Energy

Note: 'Other renewables' includes geothermal, biomass and waste energy.

ELECTRICITY PRODUCTION BY SOURCE (1 Terra Watt=10**6 watts



WORLD ELECTRICITY

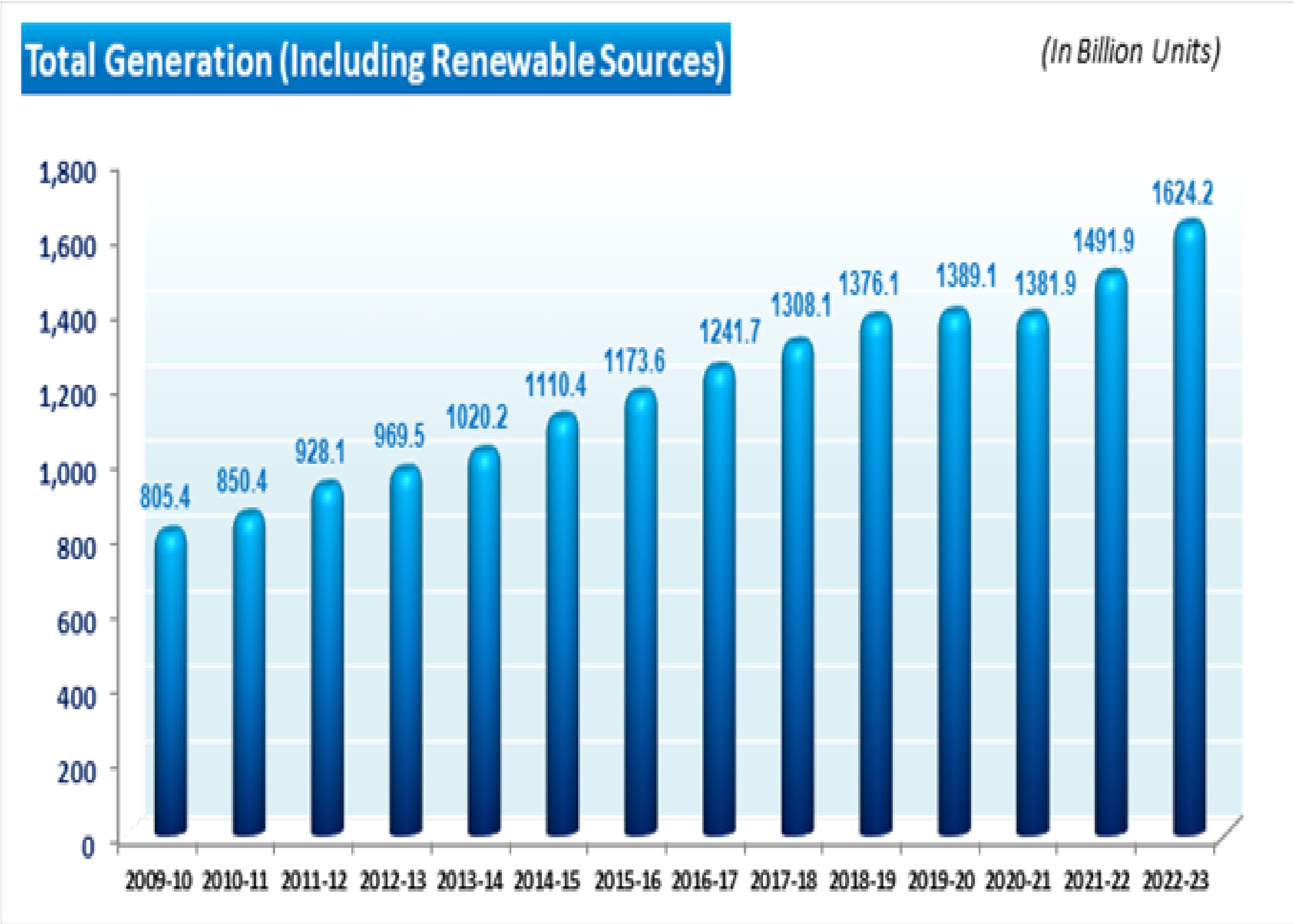


Power Sector at a Glance ALL INDIA

		Updated on 12-06-2023 **Source: OM SECTION**
1.Total Installed Capacity (As on 31.05.2023) **-** Source : Central Electricity Authority (CEA)		
INSTALLED GENERATION CAPACITY (SECTOR WISE) AS ON 31.05.2023		
Sector	**MW**	**% of Total**
Central Sector	1,00,055	24.0%
State Sector	1,05,726	25.3%
Private Sector	2,11,887	50.7%
Total	**4,17,668**	

Installed GENERATION CAPACITY(FUELWISE) AS ON 31.05.2023		
CATAGORY	**INSTALLED GENERATION CAPACITY(MW)**	**% of SHARE IN Total**
Fossil Fuel		
Coal	205,235	49.1%
Lignite	6,620	1.6%
Gas	24,824	6.0%
Diesel	589	0.1%
Total Fossil Fuel	**2,37,269**	**56.8 %**
Non-Fossil Fuel		
RES (Incl. Hydro)	**173,619**	41.4%
Hydro	**46,850**	**11.2 %**
Wind, Solar & Other RE	125,692	30.2 %
Wind	42,868	10.3 %
Solar	67,078	16.1 %
BM Power/Cogen	10,248	2.5 %
Waste to Energy	554	0.1 %
Small Hydro Power	4,944	1.2 %
Nuclear	6,780	1.6%
Total Non-Fossil Fuel	**179,322**	43.0%
Total Installed Capacity (Fossil Fuel & Non-Fossil Fuel)	**4,17,668**	**100%**

Fossil Fuel	Coal		2,06,195		48.6%
	Lignite		6,620		1.6%
	Gas		25,038		5.9%
	Diesel		589		0.1%
	Total Fossil Fuel :		**2,38,443**		**56.2%**
Non-Fossil Fuel	RES (Incl. Hydro)		1,78,365		42.0%
	Hydro	46,850		11.0%	
	Wind, Solar & Other RE	1,31,515		31.0%	
	Wind	*44,090*		*10.4%*	
	Solar	*71,610*		*16.9%*	
	BM Power/Cogen.	*10,262*		*2.4%*	
	Waste to Energy	*570*		*0.1%*	
	Small Hydro Power	*4,983*		*1.2%*	
	Nuclear		7,480		1.8%
	Total Non-Fossil Fuel :		**1,85,845**		**43.8%**
	Total Installed Capacity (Fossil Fuel & Non-Fossil Fuel)		**4,24,288**		**100%**


Total Generation (Including Renewable Sources)
(In Billion Units)
1,800
1,600
1,400
1,200
1,000
800
600
400
200
0
805.4
850.4
928.1
969.5
1020.2
1110.4
1173.6
1241.7
1308.1
1376.1
1389.1
1381.9
1491.9
1624.2
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
2015-16
2016-17
2017-18
2018-19
2019-20
2020-21
2021-22
2022-23

The power supply position in the country during 2009-10 to 2023-24

Year	Energy				Peak			
	Requirement	Availability	Surplus(+)/Defičts(-)		Peak Demand	Peak Met	Surplus(+) / Defičts(-)	
	(MU)	(MU)	(MU)	(%)	(MW)	(MW)	(MW)	(%)
2009-10	8,30,594	7,46,644	-83,950	-10.1	1,19,166	1,04,009	-15,157	-12.7
2010-11	8,61,591	7,88,355	-73,236	-8.5	1,22,287	1,10,256	-12,031	-9.8
2011-12	9,37,199	8,57,886	-79,313	-8.5	1,30,006	1,16,191	-13,815	-10.6
2012-13	9,95,557	9,08,652	-86,905	-8.7	1,35,453	1,23,294	-12,159	-9.0
2013-14	10,02,257	9,59,829	-42,428	-4.2	1,35,918	1,29,815	-6,103	-4.5
2014-15	10,68,923	10,30,785	-38,138	-3.6	1,48,166	1,41,160	-7,006	-4.7
2015-16	11,14,408	10,90,850	-23,558	-2.1	1,53,366	1,48,463	-4,903	-3.2
2016-17	11,42,929	11,35,334	-7,595	-0.7	1,59,542	1,56,934	-2,608	-1.6
2017-18	12,13,326	12,04,697	-8,629	-0.7	1,64,066	1,60,752	-3,314	-2.0
2018-19	12,74,595	12,67,526	-7,070	-0.6	1,77,022	1,75,528	-1,494	-0.8
2019-20	12,91,010	12,84,444	-6,566	-0.5	1,83,804	1,82,533	-1,271	-0.7
2020-21	12,75,534	12,70,663	-4,871	-0.4	1,90,198	1,89,395	-802	-0.4
2021-22	13,79,812	13,74,024	-5,787	-0.4	2,03,014	2,00,539	-2,475	-1.2
2022-23	15,11,847	15,04,264	-7,583	-0.5	2,15,888	2,07,231	-8,657	-4.0
2023-24	2,66,951	2,66,360	-591	-0.2	2,21,370	2,21,347	-23	-0.01

* Upto May 2023 (Provisional), Source : CEA

Installed Capacity in MW as on 01.12.2019.

Sl No	Name of Power station	Capacity, unit wise (MW)	Installed Capacity, MW
1	**KSEBL**		
1.a	**Hydro**		
1	Kuttiadi + KES+KAES	3x25+ 1x50+2x50	225.00
2	Sholayar	3x18	54.00
3	Poringalkuthu	4x9	36.00
4	PLBE	1x16	16.00
5	Pallivasal	3x5+3x7.5	37.50
6	Sengulam	4x12	48.00
7	Panniar	2x16	32.00
8	Neriamangalam+NES	3x17.5+25	77.50
9	Sabarigiri	55x4+60x2	340.00
10	Idukki	6x130	780.00
11	Idamalayar	2x37.5	75.00
12	Lower Periyar	3x60	180.00
13	Kakkad	2x25	50.00
14	Kallada	2x7.5	15.00
15	Peppara	1x3	3.00
16	Madupetty	1x2	2.00
17	Malampuzha	1x2.5	2.50
18	Chembukadavu stage I & II	0.9x3+1.25x3	6.45
19	Urumi stage I & II	3x1.25 + 3x0.8	6.15
20	Malankara	3 x 3.5	10.50
21	Lower Meenmutty	2x1.5+0.5	3.50
22	Kuttiady tail race	3x1.25	3.75
23	Poozhithode	3x1.6	4.80
24	Ranni-Perunadu	2x2	4.00
25	Peechi	1x1.25	1.25
26	Vilangad	3x2.5	7.50
27	Chimini	2.5 x 1	2.50
28	Addiyan para	2x1.5 +1x 0.5	3.50
29	Barapole	3x5	15.00
30	Vellathuval SHEP	2x1.8	3.60
31	Peruthenaruvi SHEP	2x3	6.00
		Total Hydro	**2052.00**
1.b.	**Diesel/LSHS**		
1	BDPP	3x21.32	63.96
2	KDPP	6x16	96.00
		Total Diesel	**159.96**
1.c	**Wind**		
1	Kanjikode	9x0.225	2.025
		Total Wind	**2.025**
1.d	**Solar**		
1	KSEBL solar PV kanjikode	1.0MWp	**1.00**
2	KSEBL solar PV Edayar	1.25MWp	**1.25**
3	KSEBL solar PV kollengod	1.0MWp	**1.00**
4	Barapole solar canal top	3MWp	**3.00**
5	Barapole solar canal bank	1MWp	**1.00**
6	Muvattupuzha	1.25MWp	**1.25**
7	Pothencode	2MWp	**2.00**
		Total solar	**10.50**
		Total KSEBL Generation	**2224.49**
2	**CAPTIVE**		
2.a	**Hydro**		
1	Maniyar	3x4	12.00
2	Kuthungal	3x7	21.00
		Total Hydro CPP	**33.00**
2.b	**Solar**		
1	CIAL	13MWp	29.03
2	Hindalco	1MWp	1.00
3	KMRL	2.67MWp	2.67
		Total Solar CPP	**32.70**
2.c	**Thermal**		
1	PCBL	1x10	10.00
		Total Thermal CPP	**10.00**
2.d	**Wind**		
1	**Malayala Manorama**	**10**	**10.00**
		Total wind CPP	**10.00**
3	**IPP**		
3.a	**Thermal ***		
1	RGCCPP	2x116.6 + 1x126.38	359.58
		Total IPP-Thermal	**359.58**
3.b	**Hydro**		
1	Ullunkal	2x3.5	7.00
2	Iruttukkanam	1.5x3	4.50
3	Karikkayam	3x3.5	10.50
4	Meenvallom	1x3	3.00
5	Pathankayam	2x3.5+1x1	8.00
		Total IPP- Hydro	**33.00**

Kerala Energy Scenario

Total Installed Capacity

KSEB-2823MW

Centre -359 MW

Private-527 MW

Total-3709

Peak Load Requirement-4568MW

In 2021-22

Energy Produced -10516 MU

Where do we get this Energy?

There are only two sources of energy

1.Solar Energy-1300w/sqm-5.5 million Exajoules [EJ/y](EJ=10**18 Joules)

2.Gravitational Energy-2.24 · 10 **32J

3.Nuclear Fuels-Uranium-235 , Plutonium-239 etc